

A few popular open source tools used for infrastructure configuration management in the Windows environment are Jenkins, Chef, Puppet,

Ansible, Vagrant and SaltStack. These tools differ in their simplicity and power; for example, Ansible is agentless, which makes it easy to deploy.

Continuous integration, continuous deployment and OSS tools

Continuous integration allows software developers to continuously merge their changes to a centralised repository. Once code changes are committed to the repository, continuous integration takes over the process of building and testing the new code. Hence, every small change committed to the centralised repository maintains its own integrity.

Continuous deployment is a continuation of continuous integration, wherein the tested code is put into production seamlessly through a pipeline of functions. The background activity of setting up environments for Development, QA, Staging, UAT and Production is taken care of by configuration orchestration tools that are designed to automate the deployment of servers and other infrastructure.

There are many Windows open source tools used for the suite of software development activities that encompass continuous development, continuous integration, continuous testing, continuous deployment, continuous security, and continuous monitoring and logging. Examples of these open source tools for continuous software delivery include Java for programming, MySQL for database, Tomcat for Web services, Eclipse as IDE, Maven or Ant for software builds, GIT or Bit bucket for version control, and Selenium or JUnit for automated testing. Continuous integration and deployment open source tools such as Jenkins are available as plugins and make a ton of integrations possible.



Figure 1: The DevOps culture

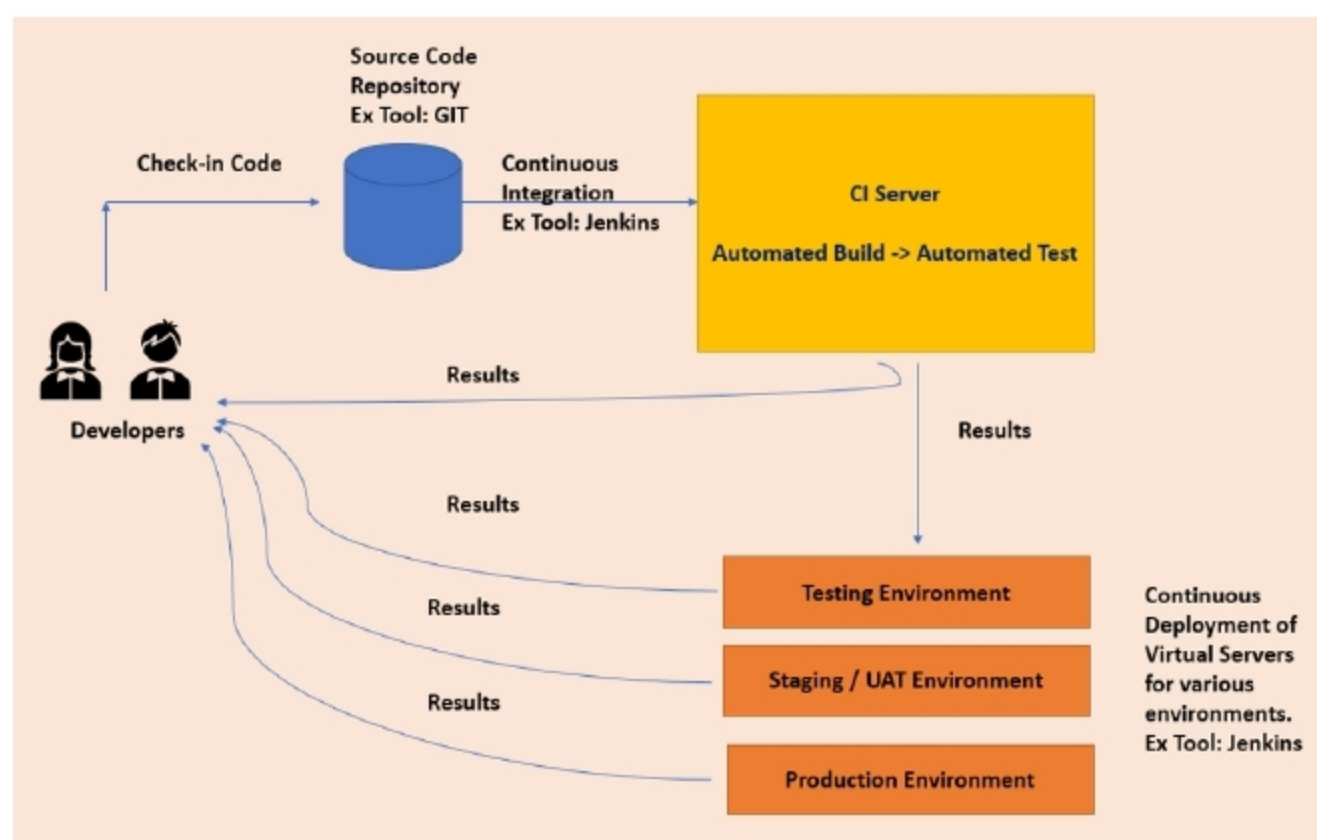


Figure 2: Continuous software integration and deployment